

Q NO:7

Question description

You are given an array A of N integers. Now, two functions $F(X)$ and $G(X)$ are defined:

$F(X)$: This is the smallest number Z such that $X < Z \leq N$ and $A[X] < A[Z]$

$G(X)$: This is the smallest number Z such that $X < Z \leq N$ and $A[X] > A[Z]$

Now, you need to find for each index i of this array $G(F(i))$, where $1 \leq i \leq N$. If such a number does not exist, for particular index i , output -1 as its answer. If such a number does exist, output $A[G(F(i))]$

Constraints:

$$1 \leq N \leq 3000$$

$$0 \leq A[i] \leq 10^{18}$$

Input :

The first line contains a single integer N denoting the size of array A . Each of the next N lines contains a single integer, where the integer on the i^{th} line denotes $A[i]$.

Output :

Print N space-separated integers on a single line, where the i^{th} integer denotes $A[G(F(i))]$ or -1 , if $G(F(i))$ does not exist.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
8
3 7 1 7 8 4 5 2
```

EXPECTED OUTPUT

```
1 4 4 4 -1 2 -1 -1
```

Test Case 2

INPUT (STDIN)

```
8
31 71 11 17 81 14 51 21
```

EXPECTED OUTPUT

```
11 14 14 14 -1 21 -1 -1
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
if(a[j]>a[i])
```

Test Case 2

KEYWORD

```
for(i=0;i<n;i++)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

9

Test Case 2

TOKEN COUNT

225

Test Case 3

NLOC

30

Code:

```
#include <stdio.h>

int main()
{
    int N;
    scanf("%d", &N);
```

```
long long A[N];

for (int i = 0; i < N; i++)
{
scanf("%lld", &A[i]);
}

for (int i = 0; i < N; i++)
{
int F = -1;
int G = -1;

for (int j = i + 1; j < N; j++)
{
if (A[i] < A[j])
{
F = j;
break;
}
}
if (F == -1)
{
printf("-1 ");
continue;
}

for (int j = F + 1; j < N; j++)
{
if (A[F] > A[j])
{
G = j;
break;
}
}

if (G == -1)
{
printf("-1 ");
}
else
{
printf("%lld ", A[G]);
}
}

return 0;
}
```

Output:

Test_case 1:

```
8
3 7 1 7 8 4 5 2
1 4 4 4 -1 2 -1 -1 %
```

Test_case 2:

```
8
31 71 11 17 81 14 51 21
11 14 14 14 -1 21 -1 -1 %
```